

HR Analytics Dashboard Project Report

Project Title

HR Analytics & Workforce Attrition Dashboard Using Tableau

Project Overview

This project analyzes workforce demographics, employee attrition patterns, and job satisfaction levels using an interactive Tableau Dashboard. The analysis investigates 1,470 employee records across organizational departments, academic fields, gender distributions, and tenure brackets. By consolidating raw human resources metrics into clear KPI cards, distribution histograms, and cross-segmented donut charts, this dashboard provides strategic human capital insights to curb voluntary turnover.

Key Performance Indicators (KPIs)

- Total Employee Count: 1,470 employees across all departments.
- Attrition Count: 237 employees have left the organization.
- Active Employees: 1,233 current active workforce members.
- Overall Attrition Rate: 16.12% company-wide turnover rate.
- Average Employee Age: 37 years old across the workforce.
- Attrition by Gender Split: 150 Male exits (63.3%) vs. 87 Female exits (36.7%).

Dashboard Visualizations & Metric Analysis

1. Department-wise Attrition Distribution

- Research & Development (R&D): 133 exits, accounting for 56.12% of total attrition.
- Sales: 92 exits, accounting for 38.82% of total attrition.
- Human Resources: 12 exits, accounting for 5.06% of total attrition.

2. Workforce Age Distribution (Bin Size: 3 Years)

- The highest employee concentration lies in the 30-36 age bracket (peaking at age 33-35 with 213 employees).
- A sharp drop in workforce headcount occurs after age 42, tapering to only 5 employees aged 60+.
- Early career headcount (ages 18-24) totals 71 employees across the organization.

3. Education Field-wise Attrition

- Life Sciences: Highest absolute attrition with 89 departures.
- Medical: Second highest departure volume with 63 exits.
- Marketing: 35 departures.
- Technical Degree: 32 departures.
- Other Fields: 11 departures.
- Human Resources: Lowest volume with 7 departures.

4. Attrition Rate by Gender Across Age Groups

- Age 25-34 Cohort: Highest vulnerability, with 69 Male exits (29.11%) and 43 Female exits (18.14%).
- Age 35-44 Cohort: Second largest group, with 37 Male exits (15.61%) and 14 Female exits (5.91%).
- Under 25 Cohort: 20 Male exits (8.44%) and 18 Female exits (7.59%).
- Age 45-54 Cohort: 16 Male exits (6.75%) and 9 Female exits (3.80%).
- Over 55 Cohort: 8 Male exits (3.38%) and 3 Female exits (1.27%).

5. Job Satisfaction Rating Matrix (Scale 1 to 4)

- Total satisfaction distribution: Level 1 (289), Level 2 (280), Level 3 (442), Level 4 (459).
- Sales Executives comprise the largest role (326 employees), with 69 reporting Level 1 satisfaction.
- Research Scientists (292 employees) and Laboratory Technicians (259 employees) report 54 and 56 employees at Level 1 satisfaction respectively.
- Managers and Research Directors demonstrate the highest percentage of Level 3 & 4 satisfaction ratings.

Key Business Insights

- Primary Demographic Flight Risk: Employees aged 25-34 represent nearly half (47.25%) of all company turnover.
- Departmental Burden: R&D and Sales combined represent 94.94% of all employee exits, requiring targeted organizational retention audits.
- STEM Background Concentration: Life Sciences and Medical graduates make up 64.1% of total departures.
- Gender Asymmetry: Male employees exit at nearly double the frequency of female employees across almost every age tier.
- Satisfaction Impact: Over 38% of the workforce rates job satisfaction at Level 1 or 2, highlighting significant workplace dissatisfaction risks in frontline technical and sales roles.

Strategic HR Recommendations

- Deploy Early-Career Retention Programs: Implement mentorship, skill-tracking, and career-progression milestones specifically targeted at the 25-34 age demographic.
- Frontline Workload Audit: Address satisfaction bottlenecks in high-churn technical roles (Laboratory Technicians and Research Scientists).
- Sales Incentive Realignment: Review compensation commission structures and target setting for Sales Executives to mitigate high sales departures.
- Regular Pulse Surveys: Implement quarterly satisfaction tracking to address Level 1 and Level 2 detractors before voluntary resignations occur.

Skills Demonstrated

- Tableau Dashboard Design & Dynamic Layout Formatting
- Cross-Filtering & Global Parameter Implementation (Education Filter, Bin Size Slicer)
- Workforce Analytics & Key Performance Indicator (KPI) Formulation
- Matrix Grid Visualizations (Job Satisfaction Heatmaps)
- Executive Storytelling & Business Recommendation Formulation

Conclusion

The Tableau HR Analytics Dashboard demonstrates how interactive business intelligence can diagnose multi-factor attrition trends. By identifying the 25-34 age cohort, R&D and Sales departments, and frontline technical positions as high-risk zones, HR leadership can move from reactive hiring to proactive, targeted employee retention.